

Mechanical drivetrain losses of a pumping-kite ground station: the TU Delft GS1 station, component by component, and where the energy goes

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Abstract

Between the tether and the motor shaft, every element of a ground station takes a share of the power: the guiding pulleys, the drum, the belt, the gearbox, and the motor's own bearings and fan. This note explains the physics of each element and how its loss scales, using the first TU Delft ground station (GS1, 2010–2011) as the worked example. It estimates GS1's mechanical losses from its hardware data on optimised LEI-V3 pumping cycles, describes the tow test behind its load-independent friction, and uses GS1's measured energy balance to show where the energy of a pumping cycle goes and what could be recovered cheaply.

Key points

- On an optimised LEI-V3 pumping cycle, GS1's drivetrain between tether and motor shaft would cost 12.1–20.4 % of the tether energy (0.81–1.37 kW of 6.72 kW). This is a hypothetical load: GS1 was rated for 4 kN, the cycle reels out at about 7.4 kN.
- 37–63 % of that is the *load-independent* friction of seals, oil, bearings and fan. The belt and the gear mesh come next; the pulleys and the drum are small.
- The load-independent friction does not shrink with the load, so the efficiency is lowest at low tension and high speed: the reel-in carries 34–44 % of the loss in about a third of the cycle.
- Optimising the cycle for shaft power instead of tether power slows the reel-in and lowers its tension, but gains only +0.5 to 1.1 % at the shaft: the drive loss is mostly a cost of the hardware, not of the flight strategy.
- That friction was measured once, in a tow test on a cold day. A warm gearbox probably has less.
- On GS1 only 39 % of the reel-out tether energy reached the electrical output. In order of size the losses were the electrical conversion, the reel-in work, friction and the auxiliaries. Cheap fixes (brakes, spindle drive, oil and seals, reel-in strategy) add about a third to the net output; a better machine and inverter add the most.

1 The power chain and how losses scale

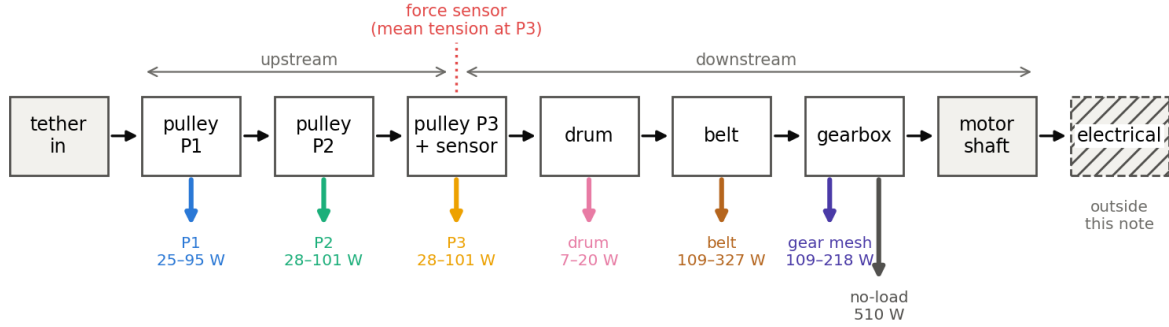


Figure 1: The mechanical power chain of GS1 from the tether to the motor shaft, with the cycle-mean loss of each element on an optimised LEI-V3 cycle (low–high estimate, Sect. 4). The force sensor on P3 splits the chain into an upstream and a downstream part (Sect. 2). The electrical conversion is not part of the estimate.

The tether delivers the power Tv , with T the tension and v the reel speed (positive in reel-out). Each element i of the chain in Fig. 1 takes a share of it, so the motor shaft receives

$$P_{\text{shaft}} = Tv - \sum_i F_i(v, T) |v|. \quad (1)$$

Each loss is written as an equivalent force $F_i \geq 0$ at the tether. This makes the elements directly comparable, whatever shaft they sit on:

- a loss torque M on a shaft of radius R that turns with the tether (a pulley, the drum) is $F = M/R$;
- a loss torque on the motor side is $F = M n / R_{\text{drum}}$, with n the overall ratio of belt and gearbox. On GS1, $n = 6.2$ and $R_{\text{drum}} = 0.1615\text{ m}$, so a modest 1 N m at the motor is already 38 N at the tether.

A loss costs energy in both reel directions. In reel-out it reduces what reaches the generator; in reel-in the motor supplies it on top of the work done on the tether.

Three scaling classes. Every mechanism below falls into one of three classes (Fig. 2):

- **Constant (Coulomb):** a fixed force, so the lost power grows with $|v|$. Contact seals, bearing preload, belt pretension.
- **Proportional to tension:** $F = kT$, a fixed fraction of the transmitted power. Bearings under the tether load, rope bending over sheaves, belt and gear-mesh friction.
- **Speed-dependent (viscous):** $F \propto |v|$ or faster, so the lost power grows at least as v^2 . Oil churning, fan windage.

The first and third classes do not depend on the load: they are paid in full even when the transmitted power is small. That is why they dominate the reel-in, where the tension is low and the speed high.

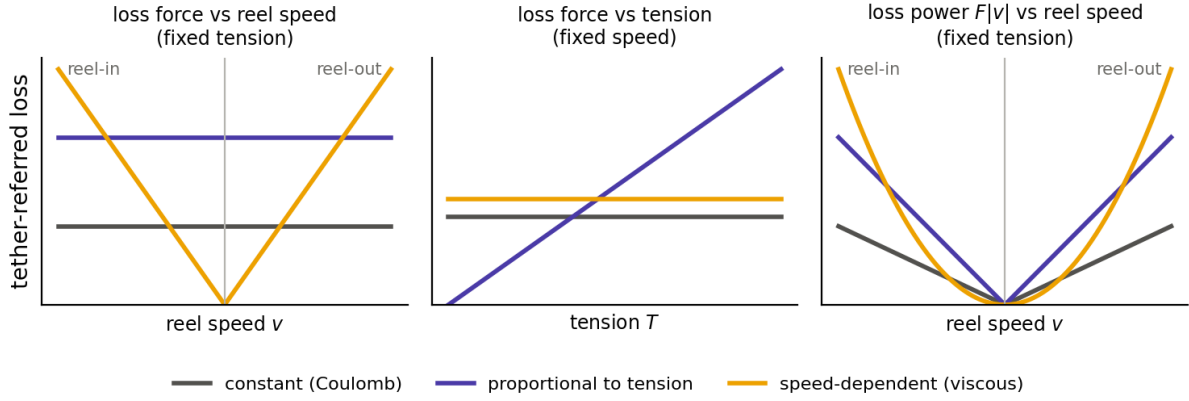


Figure 2: The three loss classes, qualitatively. Left: force against reel speed. Centre: force against tension. Right: lost power against reel speed. Only the tension-proportional class shrinks when the load is low.

Inertia (drum, gears, rotor) stores and returns energy and integrates to zero over a closed cycle. It matters for torque peaks and control, not for the cycle energy, and is not a loss here.

2 GS1: tether path and where the force is measured

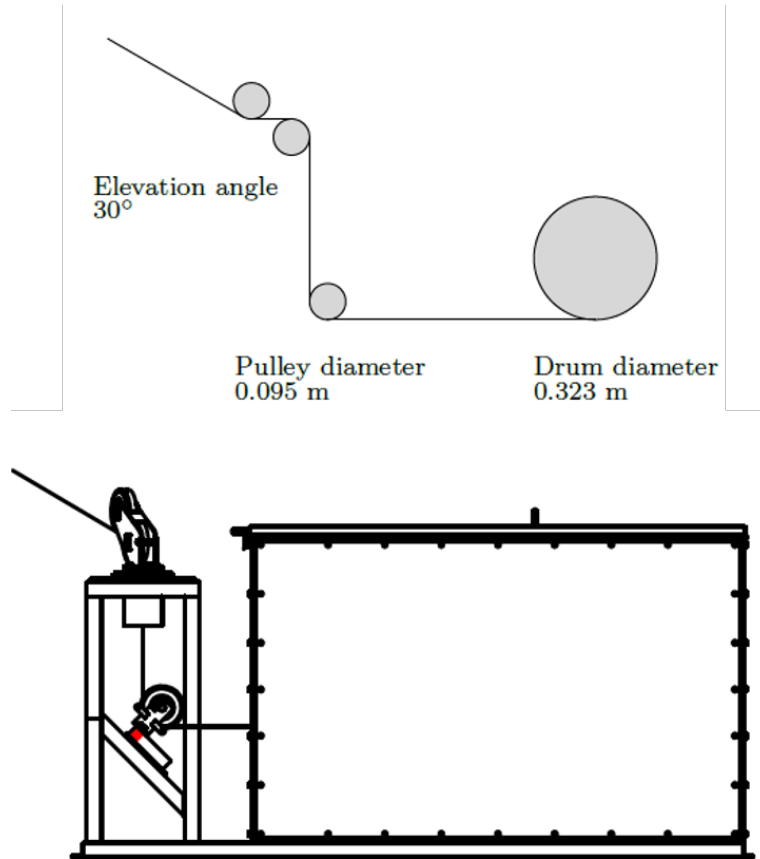


Figure 3: Tether path of GS1 (schematic above, drawing below). Pulleys P1, P2 and P3 are 0.095 m in diameter; the drum is 0.323 m. The force sensor is on the axle of P3, the lowest pulley.

GS1 [1] had one 18 kW motor-generator for both reel directions, driving the drum through a synchronous belt and a gearbox; three pulleys with a force sensor; a spindle drive for level

winding (Sect. 3.2); and a 350 V battery. The tether arrives at the elevation angle β , turns over P1 by β (30° in the drawing), over P2 by 90° to vertical, over P3 by 90° to horizontal, and runs straight onto the drum.

What the sensor reads. A pulley axle is loaded by the two rope tensions acting on it (Fig. 4). With equal tensions and a wrap angle θ the resultant is

$$F_{\text{axle}} = 2T \sin(\theta/2), \quad (2)$$

so $\sqrt{2}T$ on P3. The station's processing converts this axle force to tether tension, so the logged ground force is already the tension. What matters here is *which* tension: with friction in P3, the incoming and outgoing tensions differ by that pulley's loss. The axle resultant is $\sqrt{T_{\text{in}}^2 + T_{\text{out}}^2} \approx \sqrt{2}\bar{T}$, so the sensor reads the mean tension at P3, and half of P3's loss lies on either side of the measurement.

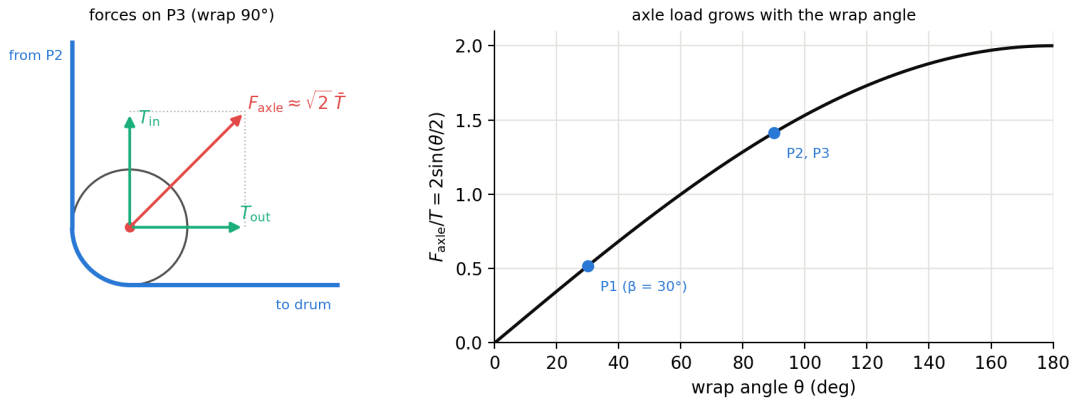


Figure 4: Left: the two rope tensions on P3 and their resultant, the force the sensor measures. Right: the axle load relative to the tension against the wrap angle. It also sets the bearing load of each pulley (Sect. 3.1).

This splits the drivetrain into two parts:

- **Upstream** of the measurement: P1, P2 and half of P3. These losses lower the tension before it is measured. A kite model calibrated on the measured ground force already absorbs them, and a comparison of sensor force with drive power does not see them.
- **Downstream:** half of P3, the drum, the belt, the gearbox and the motor's mechanical losses. This is the part a power comparison at the sensor measures.

3 Component physics

3.1 Pulleys

A pulley loses power in two ways (Fig. 5).

Bearing friction. The sheave turns on rolling bearings that carry the axle load of Eq. (2). To first order, the friction torque of a rolling bearing is $M = \mu_b F_{\text{axle}} d_b/2$, with μ_b the friction coefficient and d_b the bore diameter. Catalogues give $\mu_b \approx 0.0015$ for deep-groove ball bearings and slightly more for roller bearings; contact seals add a roughly constant torque [3]. Referred to the tether over a sheave of diameter D ,

$$F_{\text{bearing}} = \mu_b 2T \sin(\theta/2) \frac{d_b}{D}. \quad (3)$$

It grows with the tension and the wrap angle, but the small lever arm d_b/D keeps it at about 0.05 % of T per pulley.

Bending of the tether. The larger loss is in the tether itself. A rope running onto a sheave is bent to the radius $D/2$ and straightened again when it leaves. In a braided fibre rope such as a Dyneema tether, bending makes the strands slide against each other under the radial pressure the tension produces, and the fibres and coating dissipate part of the bending work (hysteresis). Both grow with the tension and with the curvature $1/D$, so per bend–unbend cycle

$$F_{\text{bend}} \approx c_b \frac{d}{D} T, \quad (4)$$

with d the rope diameter and c_b an empirical coefficient. The ratio D/d is the main design parameter for both the bending loss and the bending fatigue of ropes [4]. Each sheave costs one full cycle, whatever its wrap angle, as long as the rope conforms to the groove. With a 4 mm tether, $D/d = 24$; a coefficient of 0.05–0.2 gives 0.2–0.8 % of the tension per pulley, the typical sheave efficiency range for fibre rope. This is the dominant pulley loss and the most uncertain pulley parameter.

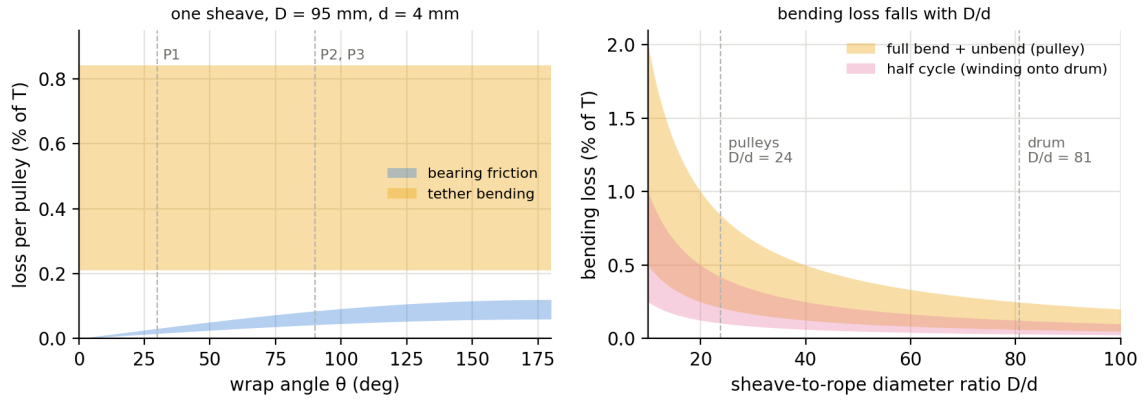


Figure 5: Left: loss of one GS1 pulley as a fraction of the tension, bearing friction against tether bending (bands = assumed parameter range). Bending dominates and does not depend on the wrap angle. Right: the bending loss falls as d/D ; the drum, with a larger diameter and only half a cycle per pass, loses far less than a pulley.

3.2 Drum and level winding

- **Bending onto the drum.** The tether is bent once when it winds on and straightened only when it pays out again: half a cycle per pass, at $D/d = 81$. Far smaller than on a pulley (Fig. 5, right).
- **Drum bearings.** They carry roughly the tension plus the drum’s weight. With journals of about a fifth of the drum diameter they cost $\mu_b T d_b/D$, a few hundredths of a percent.
- **Spooling.** With several layers, the rope presses into the layers below and rubs at the cross-overs.
- **Level winding.** To lay the layers evenly, the tether must move along the drum axis by one tether diameter per drum revolution and reverse at the drum ends. On GS1 this was done the other way round: the tether exit stayed fixed and a spindle motor, through a gearbox and a worm drive, moved the drum *together with the generator* back and forth [1]. That moves the heaviest parts of the station instead of a light guide pulley. It costs electrical power (200–600 W on GS1, Sect. 6) rather than drum torque, so it is an auxiliary load, outside the mechanical chain estimated here. Later stations solved this.

3.3 Belt

GS1 coupled the drum to the gearbox through a synchronous (toothed) belt. The teeth slide slightly as they engage and leave the pulley grooves, and the belt is bent around both pulleys under load; both losses are proportional to the transmitted force. Toothed belts reach 97–99 %, so $F_{\text{belt}} = (1 - \eta_{\text{belt}}) T$, 1–3 % of the tension. The belt pretension adds bearing load and a constant drag, which is part of the load-independent term.

3.4 Gearbox

Gearbox losses are conventionally split into load-dependent and load-independent parts [5]:

- **Mesh friction** (load-dependent). The tooth flanks slide under the transmitted load. A gear stage reaches a mesh efficiency of about 98–99 %, so $F_{\text{mesh}} = (1 - \eta_{\text{mesh}}) T$, about 1–2 % of the tension.
- **Bearings** (mostly load-dependent), as for the pulleys.
- **Oil churning and splashing** (speed-dependent). Gears move through the oil bath. The loss grows with speed and with the oil viscosity, and so depends strongly on the oil temperature (Sect. 5). Over-filling raises it.
- **Shaft seals** (constant). Lip seals have a nearly constant friction torque, higher when cold.

3.5 Motor, mechanical part

Before the conversion to electrical power, the motor contributes rotor bearing friction (constant plus load-dependent) and, for a self-ventilated motor, fan windage, whose torque grows roughly with the square of the speed. Motor standards report these together as the friction and windage loss [6]. They act at motor speed, so the ratio n amplifies them at the tether.

3.6 Summary of mechanisms

Table 1: Mechanical loss mechanisms between the tether and the motor shaft of GS1.

Element	Mechanism	Scaling	Side of sensor
P1, P2	bearing friction; tether bending (strand friction, hysteresis)	$\propto T$	upstream
P3	as P1, P2	$\propto T$	half / half
Drum	winding bend (half cycle); bearings; spooling	$\propto T$	downstream
Belt	tooth engagement; bending	$\propto T$	downstream
Gearbox	mesh friction; bearings	$\propto T$	downstream
Gearbox	oil churning	$\propto v$ or faster	downstream
Belt, gearbox, motor	pretension; seals; bearing preload	constant	downstream
Motor	fan windage	$\propto v^2$ (force)	downstream

The two-parameter law used in AWETrim’s optimiser, $F = F_c + c_v|v|$ [2], describes the load-independent part only: the constant terms and a linearised speed term. The tension-proportional part (pulleys, drum, belt, gear mesh) is not in it.

4 Estimate for GS1 on optimised LEI-V3 cycles

4.1 Inputs

The load comes from AWETrim’s optimised pumping cycles for the LEI-V3 kite. It is a *hypothetical* load for GS1: the station was rated for a maximum tether force of 4000 N [2, Table 3.1], while these cycles reel out at about 7.4 kN, nearly twice that; the speeds (up to 8 m s^{-1} both ways) are

within its limits. The tension-proportional losses scale with the load and keep their share of the power. The load-independent friction does not, so at GS1's real tensions it would take a larger share than the estimate below shows.

Table 2: Inputs of the estimate. Known values are GS1's; ranges are typical values for the element type, not measurements.

Quantity	Value	Status
Pulley diameter D (P1–P3)	0.095 m	known (drawing)
Wrap angles P1 / P2 / P3	β (30°) / 90° / 90°	known (drawing)
Drum diameter	0.323 m	known (drawing)
Tether	4 mm braided Dyneema	known
Overall ratio n	6.2 (belt + gearbox)	known [2]
Maximum tether force	4000 N	known [2]
Maximum reel speed, in / out	8 / 8 m s ⁻¹	known [2]
Bearing friction μ_b	0.0015–0.003	assumed
Bore-to-diameter ratio d_b/D	0.2	assumed
Bending coefficient c_b	0.05–0.2	assumed
Belt loss $1 - \eta_{\text{belt}}$	0.01–0.03	assumed
Mesh loss $1 - \eta_{\text{mesh}}$	0.01–0.02	assumed
Load-independent F_c / c_v	122 N / 30.7 N s m ⁻¹	measured (tow test)

The load-independent part cannot be estimated reliably from first principles, because it depends on oil, seals and motor size. It comes from GS1's tow test (Sect. 5): $\tau_s = 3.18$ N m and $c_f = 0.799$ N m s m⁻¹ on the motor side [2], which with $n = 6.2$ and $R_{\text{drum}} = 0.1615$ m is 122 N + 30.7 N s m⁻¹ $|v|$ at the tether. It is one measured value, so it carries no low–high range here; its own bias (a cold gearbox) is discussed in Sect. 5. The low–high ranges below come from the load-dependent parameters only.

4.2 Loss by component

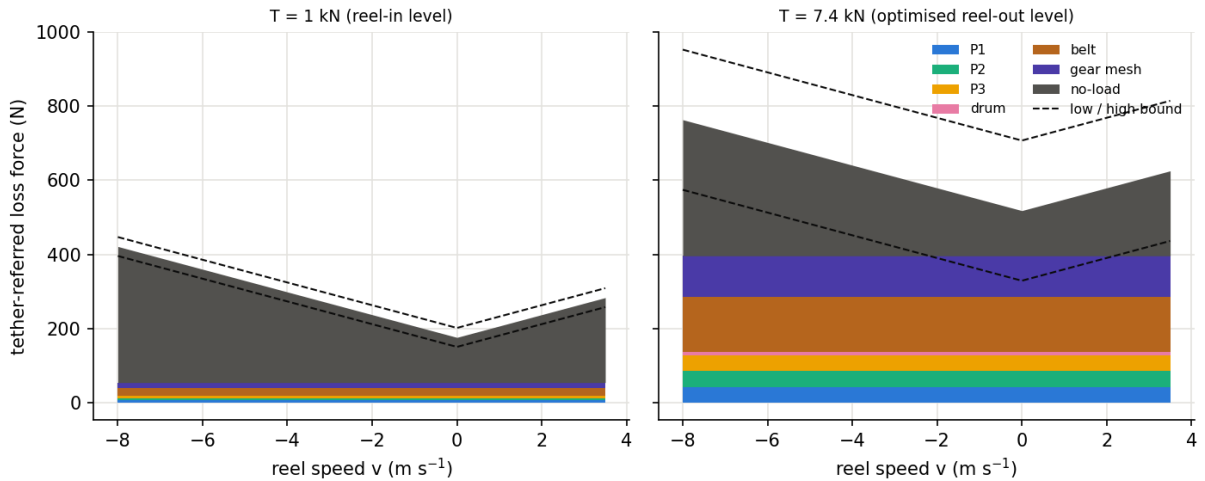


Figure 6: Tether-referred loss force by component against reel speed, at a reel-in tension (left) and an optimised reel-out tension (right). Coloured areas stack the mid-range estimates; the dashed lines are the low and high totals.

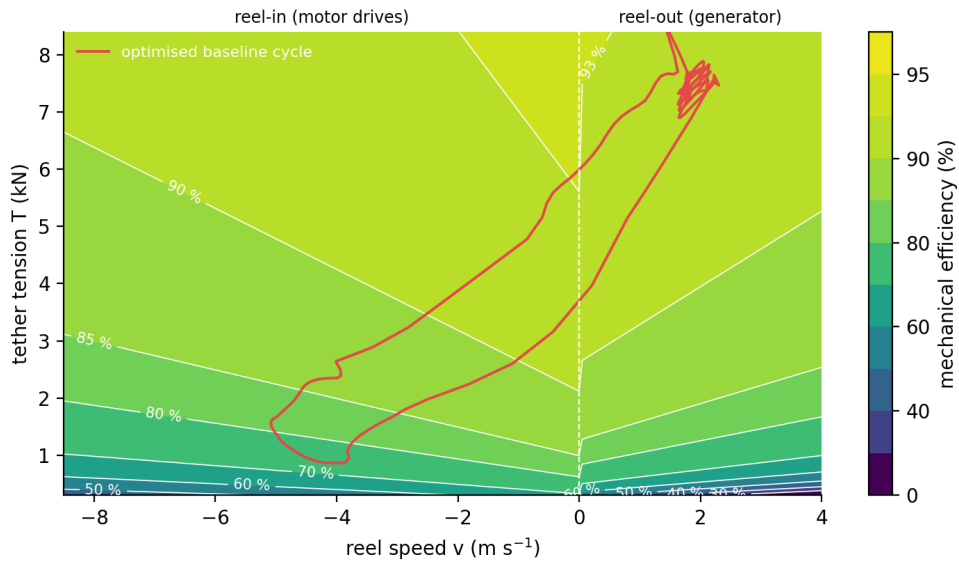
Table 3: Loss force by component (N, low–high) at representative operating points. “Upstream” and “downstream” are relative to the force sensor on P3.

Operating point	P1	P2	P3	drum	belt	gear mesh	no-load	upstream	downstream	total
Moderate reel-out (2.8 kN, $+1.10 \text{ m s}^{-1}$)	6–24	7–26	7–26	2–5	28–84	28–56	156	17–63	217–314	234–377
Optimised reel-out (7.4 kN, $+1.75 \text{ m s}^{-1}$)	17–65	19–69	19–69	5–14	74–222	74–148	176	45–168	338–594	382–761
Typical reel-in (1.0 kN, -3.50 m s^{-1})	2–9	3–9	3–9	1–2	10–30	10–20	229	6–23	251–286	257–309
Fast reel-in (1.0 kN, -8.00 m s^{-1})	2–9	3–9	3–9	1–2	10–30	10–20	368	6–23	389–424	396–447

- **At low tension**, as in reel-in, the load-independent friction is almost the entire loss, and it grows with the reel speed.
- **At high tension**, as in optimised reel-out, the tension-proportional elements become visible. At 7.4 kN: the belt 74–222 N, the gear mesh 74–148 N, the three pulleys together 54–202 N, against 176 N of load-independent friction. The drum costs a few newtons.
- **P1’s loss barely depends on elevation**: its bending loss is independent of the wrap, and only its small bearing term follows $\sin(\beta/2)$.

4.3 Efficiency map

Dividing the losses by the transmitted power gives the mechanical efficiency at every operating point (Fig. 7). In reel-out it is shaft over tether power; in reel-in, where the motor drives, tether over shaft power. The map shows the consequence of the scaling classes: at high tension the fixed friction is small next to the transmitted power and the efficiency is around 90 %; at low tension and high speed it falls well below. The optimised cycle spends its reel-out in the good corner and its reel-in in the poor one.

**Figure 7:** Mechanical drivetrain efficiency of GS1 (mid-range estimate) against reel speed and tension, with the optimised LEI-V3 baseline cycle overlaid.

4.4 Over a pumping cycle

Figure 8 shows the optimised LEI-V3 baseline cycle (logarithmic wind 10 m s^{-1} at 100 m, tether power 6.72 kW) with the loss power of each element over time.

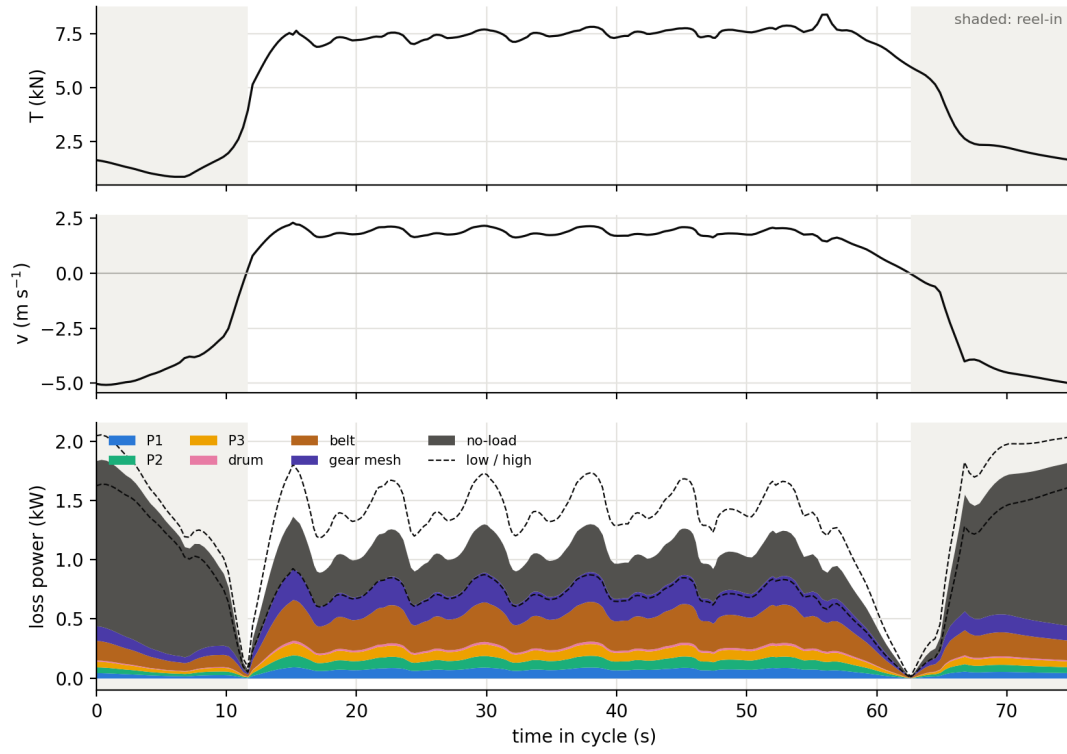


Figure 8: Tension, reel speed and mechanical loss power over the optimised baseline cycle. The stacked areas are the mid-range estimate per component, the dashed lines the low and high totals; reel-in is shaded.

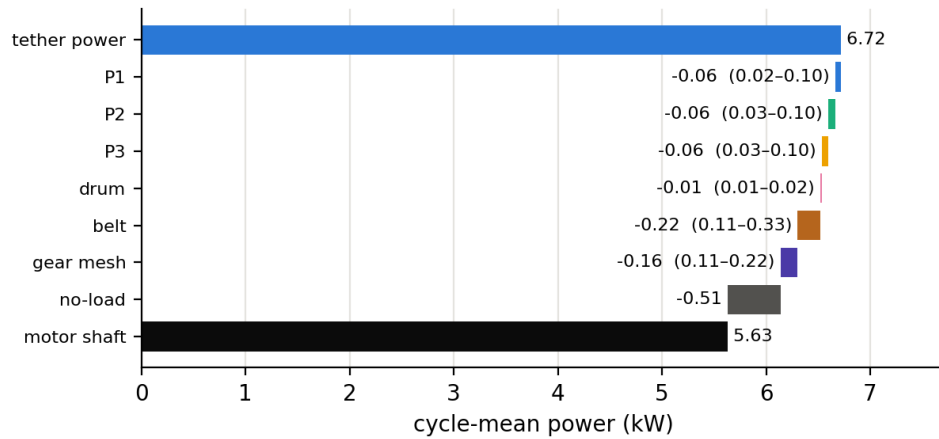


Figure 9: Cycle-mean power from the tether to the motor shaft on the baseline cycle: mid-range loss per element, low-high in brackets (kW).

Table 4: Share of the cycle's tether energy lost per component (low-high).

	P1	P2	P3	drum	belt	gear mesh	no-load	upstream	downstream	total
mean loss power (W)	25–95	28–101	28–101	7–20	109–327	109–218	510	66–246	748–1125	814–1371
share of $\int T v dt$ (%)	0.4–1.4	0.4–1.5	0.4–1.5	0.1–0.3	1.6–4.9	1.6–3.2	7.6	1.0–3.7	11.1–16.7	12.1–20.4

- **Total.** The mechanical losses take 12.1–20.4% of the tether energy, of which 11.1–16.7% is downstream of the sensor.
- **The load-independent friction** takes 7.6%, the largest single item.

- **Belt** 1.6–4.9 %, **gear mesh** 1.6–3.2 %, **the three pulleys** together 1.2–4.4 % (mostly upstream of the sensor), **the drum** almost nothing.
- **Peaks of 1.6–2.1 kW** occur at the fast start of the reel-in and in the reel-out (Fig. 8).

4.5 Power loss at operating points and across cycles

The loss power of each element is its force times the reel speed, $P_i = F_i |v|$ (Table 5).

Table 5: Loss power by component (W, low–high) at the operating points of Table 3.

Operating point	P1	P2	P3	drum	belt	gear mesh	no-load	upstream	downstream	total
Moderate reel-out (2.8 kN, +1.10 m s ⁻¹)	7–27	8–29	8–29	2–6	31–92	31–62	171	19–70	239–345	257–415
Optimised reel-out (7.4 kN, +1.75 m s ⁻¹)	29–113	33–120	33–120	8–24	130–388	130–259	308	78–293	591–1039	669–1332
Typical reel-in (1.0 kN, -3.50 m s ⁻¹)	8–31	9–32	9–32	2–6	35–105	35–70	803	21–79	880–1001	901–1080
Fast reel-in (1.0 kN, -8.00 m s ⁻¹)	18–70	20–74	20–74	5–15	80–240	80–160	2941	48–181	3116–3393	3164–3574

- **Reel-out.** At 7.4 kN and 1.75 m s⁻¹ (13 kW at the tether) the drivetrain takes 0.67–1.33 kW, or 5.2–10.3 %.
- **Reel-in.** At 3.5 m s⁻¹ and 1 kN the tether absorbs 3.5 kW and the drivetrain adds 0.90–1.08 kW.
- **Fast reel-in.** At 8 m s⁻¹ the drivetrain adds 3.16–3.57 kW, about 40 % on top of the 8 kW the tether absorbs.

Table 6: Cycle-mean mechanical loss power of GS1 on the optimised LEI-V3 cycles (low–high). “Shaft” is the mean power at the motor shaft, tether power minus the loss.

Cycle	tether (kW)	loss (kW)	loss (%)	shaft (kW)	reel-in share of loss (%)	peak loss (kW)
baseline (multi-lobe) (75 s)	6.72	0.81–1.37	12.1–20.4	5.35–5.91	34–44	1.6–2.1
up-loop eight (74 s)	6.06	0.76–1.27	12.5–20.9	4.79–5.30	34–43	1.4–1.9
symmetric eight (57 s)	4.63	0.59–1.04	12.7–22.4	3.59–4.04	33–41	0.9–1.8
symmetric eight, mirrored (79 s)	4.76	0.63–1.08	13.2–22.7	3.68–4.13	34–42	1.0–1.8

Across the four optimised cycles (Table 6) the drivetrain takes 0.59–1.37 kW of 4.6–6.7 kW, so the shaft receives 77–88 % of the tether power. The reel-in carries 33–44 % of the loss although it lasts only about a third of the cycle, because it runs about twice as fast as the reel-out. The percentage hardly depends on the cycle shape (12.1–22.7 %), so the drivetrain lowers all cycles by a similar factor and does not change their ranking.

4.6 Optimising for shaft power instead of tether power

The cycles above were optimised for the cycle-mean *tether* power Tv and the drive losses were evaluated afterwards. AWETrim’s optimiser can instead maximise the cycle-mean *shaft* power, the tether power minus the drive losses of this section, written in the same tether-referred form:

$$P_{\text{shaft}} = Tv - (F_c + kT)v \tanh(v/\varepsilon) - c_v v^2, \quad (5)$$

with the GS1 tow-test friction ($F_c = 122$ N, $c_v = 30.7$ N s m⁻¹), the tension-proportional fraction $k = 0.0535$ of the mid estimate (pulleys, drum, belt, gear mesh; range 0.028–0.079) and $\varepsilon = 0.05$ m s⁻¹ smoothing the sign of v (`sim_parameters.winch.friction` and `drivetrain.load_fraction`). Table 7 compares the optima of the two objectives on the multi-lobe baseline and the up-loop eight.

Table 7: Optima of the tether-power and shaft-power objectives, both evaluated for tether and shaft power (mid estimate, low–high in brackets). Barred values are time means over the reel-out (out) and the reel-in (in).

Cycle	objective	period (s)	tether (kW)	shaft (kW)	reel-in time (%)	$\bar{v}_{\text{out}}$ (m s^{-1})	$\bar{T}_{\text{out}}$ (kN)	$\bar{v}_{\text{in}}$ (m s^{-1})	$\bar{T}_{\text{in}}$ (kN)
baseline (multi-lobe)	tether power Tv	75	6.72	5.63 (5.35–5.91)	32	1.74	7.38	3.67	2.22
baseline (multi-lobe)	shaft power, from the Tv optimum	79	6.68	5.66 (5.39–5.92)	32	1.64	7.54	3.50	2.07
baseline (multi-lobe)	shaft power, from the seed	66	6.28	5.31 (5.05–5.57)	31	1.49	7.95	3.33	2.33
up-loop eight	tether power Tv	74	6.06	5.04 (4.79–5.30)	37	1.82	6.82	3.09	2.17
up-loop eight	shaft power, from the seed	69	6.00	5.10 (4.86–5.33)	36	1.63	7.01	2.90	1.68

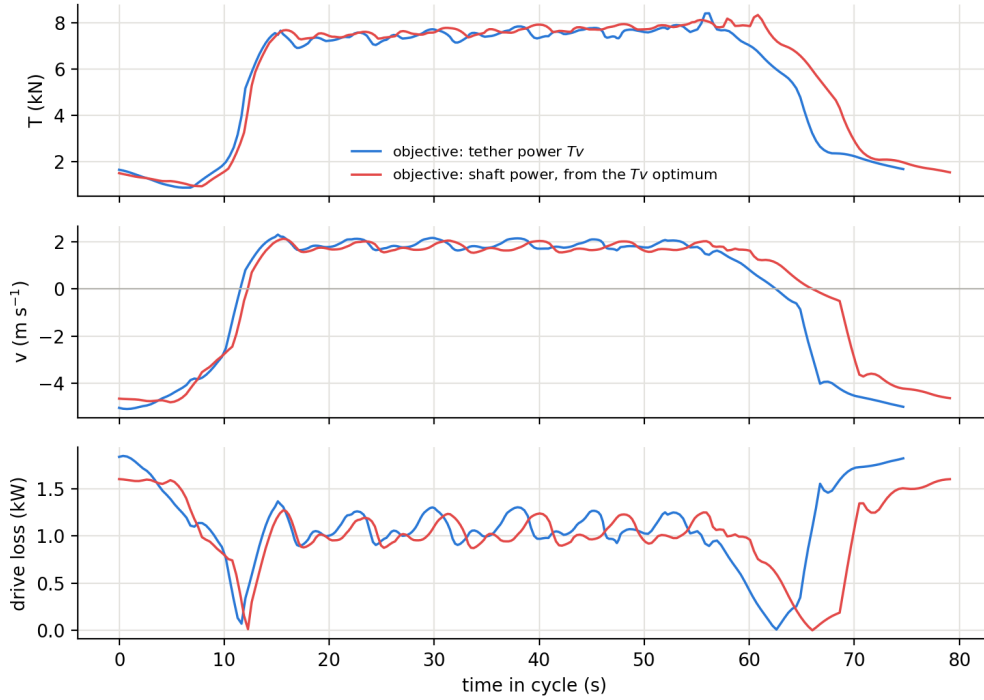


Figure 10: Multi-lobe baseline: tension, reel speed and drive loss over the cycle for the tether-power optimum (blue) and the shaft-power optimum started from it (red).

- **The gain is small.** Started from the tether-power optimum, the shaft-power optimum of the baseline delivers 5.66 kW at the shaft instead of 5.63 kW (+0.5 %), for -0.5 % tether power. The up-loop eight gains +1.1 %. The optimum is flat: the drive loss is mostly a cost of the hardware, not of the way the cycle is flown.
- **The strategy moves the way the loss classes predict.** The reel-in becomes slower (mean $3.67 \rightarrow 3.50 \text{ m s}^{-1}$, peak $5.09 \rightarrow 4.80 \text{ m s}^{-1}$ on the baseline; $3.09 \rightarrow 2.90 \text{ m s}^{-1}$ on the eight) and is flown at lower tension, so the reel-in tether work drops by -7 % and -25 %. The reel-out becomes slower and stronger ($7.38 \rightarrow 7.54 \text{ kN}$). Both changes cut the speed-dependent friction; the tension-proportional loss is a fixed fraction of the power and does not favour either.
- **A fast reel-in still pays.** The tether-power optimum already reels in well below the 8 m s^{-1} limit, trading the reel-in time against the aerodynamic work of pulling the kite back. The drive losses shift that trade-off only a little; they do not make a slow reel-in better.
- **Start from the tether-power optimum.** Started from the seed, the shaft-power optimisation of the baseline ended in another local optimum, -5.6 % below the tether-power optimum's shaft power (Table 7). A changed objective should move a known optimum, not

re-search from the seed.

These runs use GS1's losses on loads GS1 could not carry (Sect. 4), and the speed-dependent term comes from the cold tow test. The electrical conversion is not in the objective; it would add a copper loss growing with T^2 and a part-load penalty in reel-in (Sect. 6.1).

5 The measured friction: a tow test on GS1

The test. GS1's load-independent friction was measured by towing the tether with a car. The generator was disconnected electrically but stayed coupled, so its rotor turned with the drum. A force sensor at one end of the tether gave the pulling force. Repeating the pull at several steady speeds gives a straight line $F = F_c + c_v v$ (Fig. 11, left): the intercept is the sliding Coulomb friction, the slope the viscous coefficient. The test was done on a cold day. Not recorded here are the tow speeds, the temperatures, and how much tether lay on the ground during the pull.

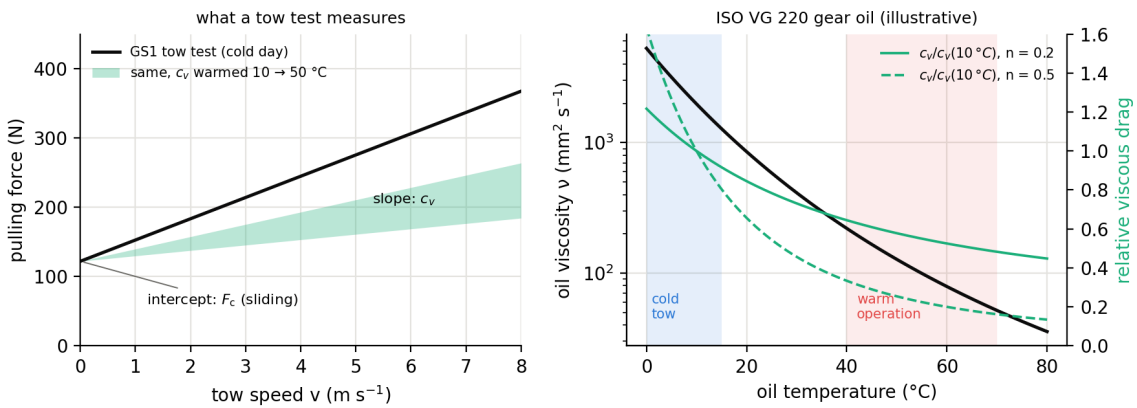


Figure 11: Left: the tow-test law and the same law with its viscous slope corrected from a cold to a warm gearbox. Right: viscosity of an ISO VG 220 gear oil against temperature (Walther equation, $220 \text{ mm}^2 \text{s}^{-1}$ at 40°C) and the resulting change of viscous drag for $c_v \propto \nu^n$. The oil grade is illustrative.

Critique. The tow test is the only measurement of this term, and it has limits:

1. **Cold oil biases it high.** Gear-oil viscosity changes very strongly with temperature: about ninefold between 10°C and 40°C for a VG 220 oil (Fig. 11, right). Churning drag grows less than linearly with viscosity, roughly as ν^n with $n = 0.2\text{--}0.5$, so a gearbox at 50°C has a viscous drag of about 25–60 % of its value at 10°C . Seal friction also falls as the seals warm. For a warm gearbox the tow test is closer to an upper value than a central one.
2. **It is a no-load test.** The tow tension is low, so the tension-proportional losses (belt, gear mesh, bearings) are absent. They come on top, as in Sect. 4.
3. **The generator is included.** Electrically disconnected but turning, the generator contributes its bearings, fan and open-circuit iron losses (hysteresis behaves like a Coulomb term, eddy currents like a viscous one). These belong to the machine's drag. If an electrical efficiency used downstream already includes iron losses, they are counted twice.
4. **Test details.** If the tow speeds stayed below the 8 m s^{-1} of fast reel-in, c_v is extrapolated there. Tether dragging on the ground would add to the measured force. A single pull to breakaway, instead of steady speeds, would give the higher static friction.

6 The measured energy chain of GS1

Fechner's AWEC 2011 talk [1] reports an energy balance of GS1, with four current sensors on the electrical side, for three consecutive pumping cycles (test 104, flight 3; wind 6 m s^{-1} at 6 m,

maximum reel-out and reel-in speeds 5.8 and 8 ms^{-1} , mean reel-out power 11.6 kW). Table 8 regroups it per reel phase, and Fig. 12 follows cycle 1 from the tether to the electrical output. The mechanical energies are sensor force times reel speed, so the friction sits downstream of the sensor, as in Sect. 2. The electrical losses include the motor controller.

Table 8: GS1 energy per cycle, regrouped per reel phase from [1] (slide 13), in Wh. “Terminals” is the electrical energy of the main machine. Efficiency is terminal over tether energy in reel-out and tether over terminal energy in reel-in.

		Cycle 1	Cycle 2	Cycle 3
Reel-out	tether energy	200	214	208
	friction	−14	−14	−15
	electrical losses	−33	−37	−33
	at the terminals	153	163	160
	efficiency	76.5 %	76.2 %	76.9 %
Reel-in	tether energy	39	44	42
	friction	+14	+18	+16
	electrical losses	+7	+8	+6
	at the terminals	60	70	64
	efficiency	65.0 %	62.9 %	65.6 %
Auxiliaries	spindle motor / brakes, computers	11 / 4	13 / 5	12 / 5
Net electrical	from the rows above	78	75	79
	reported	78	74	78

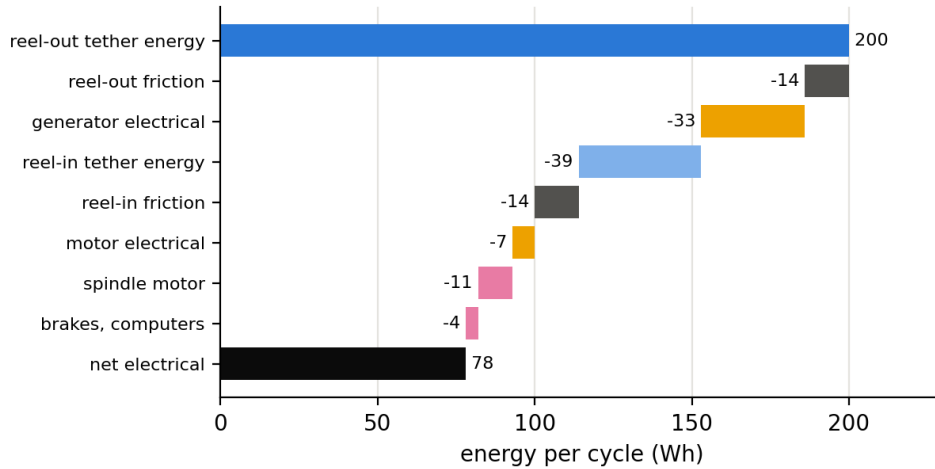


Figure 12: GS1 cycle 1 from the reel-out tether energy to the net electrical output, from [1].

- **Reel-out conversion is about 76 %:** friction takes 7 % of the tether energy, the generator and controller 16–17 %.
- **Reel-in conversion is about 65 %, and the order is reversed:** friction (14–18 Wh) is twice the electrical loss (6–8 Wh). The tension is low, so the copper losses, which grow with the square of the torque, are small, while the friction grows with the faster reel-in speed. This is the efficiency map of Fig. 7 seen in measurements.
- **Friction is 17–20 % of the net mechanical energy,** and about half of it is spent in reel-in.

Consistency check. With the talk’s crest factors (1.71 out, 1.85 in) read as maximum over mean, the mean speeds are 3.4 ms^{-1} out and 4.3 ms^{-1} in; with a period of about 115 s and a

duty cycle of 0.53 the phases last about 61 and 54 s. The tow-test law then gives $(F_c \bar{v} + c_v \bar{v}^2) t = 13.0 \text{ Wh}$ for reel-out and 16.5 Wh for reel-in, against the reported 14–15 and 14–18 Wh. The agreement is close but probably circular: the current sensors measure the machine’s total loss, and splitting it into friction and electrical loss needs a friction model, most likely this same law. Only the total (mechanical minus electrical) is a measurement, and the tension-proportional mechanical losses of Sect. 4 are then booked as electrical.

6.1 The same station in Fechner’s thesis

Fechner’s thesis [2] analyses GS1 as the “20 kW demonstrator” (Table 3.1 there: drum radius 0.1615 m, 4 mm tether, 8 m s^{-1} reel speeds, 4 kN, 20 kW nominal generator power; the talk says 18 kW). It uses the same efficiency chain under different names (Table 10): the talk’s η_m is the *pumping efficiency* η_p , $D \eta_p$ is the *cycle efficiency* η_{cyc} , and the talk’s η_e splits into a gross electrical efficiency and a system efficiency for the brakes and the spindle:

$$\eta_e^{\text{gro}} = \frac{\eta_{e,o} \eta_{e,i} \eta_{\text{bat}} - 1 + \eta_p}{\eta_{e,i} \eta_{\text{bat}} \eta_p}, \quad \eta_{\text{tot}} = D \eta_{\text{sys}} \eta_e^{\text{gro}} \eta_p, \quad (6)$$

with $\eta_{e,o}$, $\eta_{e,i}$ the generator and motor efficiencies (friction included) and η_{bat} the battery’s (Eqs. 3.26 and 3.28 there). The form makes the point of Sect. 7 explicit: because the reel-in energy passes through both machines and the battery, the gross electrical efficiency is always lower than the generator’s own.

Loss model. The thesis models the generator loss as friction plus copper loss (Eqs. 3.30–3.35 there): $P_{e,o} = P_{m,o} - 3R_g I_o^2 k - \tau_{f,o} \omega$, with the friction torque $\tau_{f,o} = \tau_c + c_{v,f} \omega$ measured on the station (the tow test of Sect. 5) and a factor k for skin effect, stray-load and other losses not modelled explicitly. The tension-proportional mechanical losses of Sect. 4 (belt, gear mesh, bearings) have no term of their own there; they would be absorbed in k .

Numbers. Table 9 compares the talk with the thesis’s measured and simulated performance factors of the same station. They agree within the spread between flights and wind conditions.

Table 9: GS1 performance factors in the AWEC 2011 talk [1] and in the thesis [2, Table 3.3 and Sect. 3.3.2].

	Talk, measured	Thesis, measured	Thesis, simulated
Pumping efficiency η_p (η_m)	79.9 %	77 %	77 %
Duty cycle D	53 %	56 %	46.8 %
Electrical efficiency	46.4 %	about 54 %	–
Total efficiency η_{tot}	19.6 %	21 %	18 %
Mean reel-out mechanical power	11.6 kW	14.1 kW	21.6 kW
Mean electrical output	2.29 kW	2.6 kW	4.0 kW

Assumed against measured machine efficiencies. For GS1 the thesis’s quasi-static model assumes a generator efficiency of 80 %, a motor efficiency of 79 % and a battery efficiency of 95 % (Sect. 3.2.8 there). The talk’s energy balance gives 76.5 % in reel-out and only 65 % in reel-in (Table 8, friction and controller included). The reel-in shortfall is what the thesis itself explains in its Sect. 3.4: a machine’s efficiency drops sharply at 10 or 20 % of its nominal power, and GS1 reeled in at about 3 kW on a 20 kW machine.

The thesis’s diagnosis. The thesis attributes GS1’s low electrical efficiency to “an asynchronous, squirrel cage generator ... together with a very inefficient gearbox and a synchronous belt” (Sect. 3.3.2 there), and proposes a synchronous direct drive. That is the order of Sect. 7: the machine first, then belt and gearbox.

7 Why the total efficiency is so low, and the low-hanging fruit

Which efficiency is which. Several efficiencies appear in this note and in [1]; Table 10 lists what each one compares. All energies are per cycle; “tether energy” is the sensor force times reel speed, integrated over a phase.

Table 10: Efficiencies used in this note. E_o , E_i : tether energy in reel-out and reel-in; $E_m = E_o - E_i$: net mechanical energy; E_e : net electrical energy, after the auxiliaries; t_o , t_i : phase durations. “Thesis”: the name in [2] (Sect. 6.1). GS1 values from [1].

Name	Thesis	Compares	GS1
Mechanical drivetrain efficiency (Fig. 7)	–	shaft over tether power at one instant (reel-in: the inverse); the mechanical chain only	–
Duty cycle D	D	$t_o/(t_o + t_i)$; a time share, not a loss	53 %
Mechanical cycle efficiency η_m	η_p	E_m/E_o : how much reel-out work is not spent on the reel-in; set by the kite and flight, not the drivetrain	79.9 %
Electrical efficiency η_e	$\eta_e^{\text{gro}} \eta_{\text{sys}}$	E_e/E_m : net electrical over net mechanical energy; includes friction, machine and controller losses in both phases, and the auxiliaries	46.4 %
Total efficiency η_{tot}	η_{tot}	$D \eta_m \eta_e$: cycle-mean electrical power over the mean reel-out mechanical power	19.6 %
Energy share	–	$E_e/E_o = \eta_m \eta_e$: net electrical over reel-out tether energy	39 %

The headline number. The talk’s 19.6 % total efficiency is $0.53 \times 0.799 \times 0.464$. The duty cycle in it is not lost energy: it says the station is idle as a generator during reel-in, which matters for the installed power and its cost, not for the energy. In energy terms, 78 Wh of 200 Wh, or 39 % of the reel-out tether energy, reached the electrical output. That is still poor.

The culprits. Of the 200 Wh reel-out tether energy of cycle 1 (Fig. 12):

1. **Electrical conversion**, 40 Wh (20 %). The largest item, mostly in reel-out (33 Wh): the generator and its controller convert about 82 % of the shaft energy. One machine serves both directions over a wide torque and speed range, and at part load its fixed losses weigh heavily.
2. **Reel-in tether work**, 39 Wh (20 %). The work done pulling the kite back. This is set by the kite and the flight strategy, not the station.
3. **Friction**, 28 Wh (14 %). Half of it in reel-in, where the speed is high and the load low.
4. **Auxiliaries**, 15 Wh (8 %). The spindle motor (200–600 W) and the brakes and computers (about 150 W) run all the time.

Reel-in energy is paid twice. Every watt-hour the motor spends in reel-in was first generated in reel-out and passed through the generator. On GS1, 1 Wh of reel-in tether work costs $1/0.65 = 1.5$ Wh at the terminals, which took $1.5/0.765 \approx 2$ Wh of reel-out tether energy. The same multiplier applies to the reel-in friction. Reducing the reel-in work and the reel-in friction therefore pays about double.

Levers. Figure 13 applies each lever alone to GS1 cycle 1. The effects are illustrative: they rest on the stated assumptions, not on measurements.

- **Brake holding current** (cheap). Spring-applied brakes draw current the whole time they are released. An economiser drops the holding current to about a third after pull-in. Assumed 4 Wh $\rightarrow$ 1 Wh per cycle: +3 Wh.
- **Spindle drive** (cheap to moderate). GS1's spindle moved the drum and the generator, the heaviest parts of the station, through a worm drive, which slides rather than rolls and typically transmits only 40–70 % of its input. Moving only a light guide pulley, with a ball or lead screw and a small motor, and only while the tether moves, needs a fraction of that. Assumed 11 Wh $\rightarrow$ 4 Wh: +7 Wh.
- **Oil and seals** (cheap). The lowest oil viscosity the gearbox allows, the correct fill level (over-filling raises churning), a warm-up before flying, and low-friction seals. Assumed viscous drag -30% and seal drag -1.5 Wh per phase: +7 Wh.
- **Reel-in strategy** (free, in software). Fly the reel-in more depowered and at a better elevation, and choose the reel-in speed with the friction in view: a faster reel-in shortens the cycle but pays friction that grows with speed. AWETrim's cycle optimiser can maximise the shaft power instead of the tether power (Sect. 4.6); on the LEI-V3 cycles that cuts the reel-in work by 7–25 % but gains only about 1 % at the shaft. Assumed reel-in tether work -25% : +10 Wh.
- **Machine and inverter** (not cheap, largest). A permanent-magnet machine with a modern inverter reaches about 94 % in each direction at these loads, against the 82 and 88 % of GS1's machine and controller: +25 Wh.
- **Drivetrain architecture** (a redesign). Remove the belt and the gearbox with a direct drive, and size the machines for their phases: a high-torque generator for reel-out and a smaller, faster motor for reel-in, so that neither runs at a few percent of its rating. A clutch decouples the generator during reel-in, so its friction is not paid there. This is the design the thesis proposes for a 53.5 kW station [2, Sect. 3.4]; with 90 % machines it simulates a gross electrical efficiency of 85–88 % and a total efficiency of 57 %. It is not in Fig. 13.

Together the four cheap levers raise the net output from 78 to about 105 Wh per cycle, a third more; with the machine upgrade it is about 131 Wh, 66 % of the reel-out tether energy instead of 39 %. The pulleys and the drum are not worth touching: they cost a few percent at most.

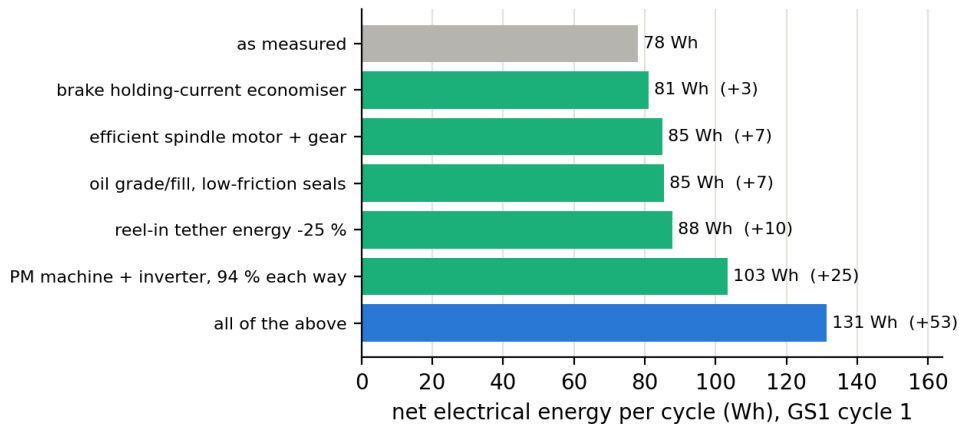


Figure 13: Net electrical energy of GS1 cycle 1 as measured and with each lever applied alone (assumptions in the text). Green: the levers; blue: all together.

8 What would pin such an estimate down

For any ground station, these measurements would replace the assumptions above:

1. **A warm no-load constant-speed test:** tether slack, motor driving at several steady

speeds in both directions up to the reel-in maximum, torque logged, after the gearbox has reached its operating temperature. This measures F_c and c_v , the dominant term, directly; repeating it with the motor decoupled separates machine and gearbox.

2. **The gearbox oil grade and its sump temperature in operation**, to correct a cold test.
3. **The gearbox and belt datasheets**: efficiency against load and speed, oil fill and seal type.
4. **Bearing types and sizes** of the pulleys and the drum.
5. **A second force measurement at the drum**, for example from the motor torque on a calibrated drive. Its difference to the sensor on the last pulley would isolate the downstream losses.
6. **An energy balance like Table 8**, with mechanical and electrical sides measured independently, including the auxiliaries.

References

- [1] U. Fechner, *Performance factors for HAWP systems in pumping operation*, presentation, Airborne Wind Energy Conference (AWEC) 2011, KU Leuven, 2011.
- [2] U. Fechner, *A Methodology for the Design of Kite-Power Control Systems*, PhD thesis, Delft University of Technology, 2016, <https://doi.org/10.4233/uuid:85efaf4c-9dce-4111-bc91-7171b9da4b77>.
- [3] SKF, *Rolling bearings* (general catalogue), section on bearing friction and the simplified friction-torque model.
- [4] K. Feyrer, *Wire Ropes: Tension, Endurance, Reliability*, 2nd ed., Springer, 2015.
- [5] ISO/TR 14179-1:2001, *Gears — Thermal capacity — Part 1: Rating gear drives with thermal equilibrium at 95 °C sump temperature*.
- [6] IEC 60034-2-1, *Rotating electrical machines — Part 2-1: Standard methods for determining losses and efficiency from tests*.